

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a71266c4fce48a38a.jsonl`  
- SHA-256: `fdcc5b64cd9844a27d1344023746dc4f0f4d445c86d2330d8b4e77987fe92ac4`  
- Source modified: `2026-06-06T17:35:17+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:34:13.420Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"The acoustic ML classifier achieves >95% detection accuracy, but the end-to-end on-court system drops to 85% average accuracy, showing real-world deployment degrades audio-only hit detection substantially."

****Source:**** <https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/> (primary)

****Supporting quote:**** "the ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:34:15.855Z)

I'll verify this claim by examining the source and searching for corroborating or contradicting evidence.

3. assistant (2026-06-06T17:34:18.668Z)

Let me fetch the primary source to examine the claim's context.

4. user (2026-06-06T17:34:28.210Z)

Summary of Tennis Timing Study

Overview

This paper investigates a **machine learning-based acoustic detection system** for analyzing temporal aspects of **tennis** performance, specifically rally rhythm and groundstroke timing.

Sport & Task

The study focuses on **tennis**. The detection task involves identifying:

- Ball impact on racket strings ("impact")
- Ball bounce on court ("rebound")

ML Algorithm Accuracy (>95%)

The exact statement about the >95% figure:

> "The ML algorithm showed a detection accuracy higher than 95%"

This refers specifically to the machine learning classifier's ability to distinguish tennis ball sounds from noise in controlled settings.

On-Court System Accuracy (85%)

The exact statement about the 85% figure:

> "the average accuracy of the whole system that was applied on-court was 85%"

What "Whole System Applied On-Court" Means

The 85% figure reflects **audio-only detection accuracy degrading in real field conditions**. The paper explains that detection accuracy dropped due to environmental noise in on-court settings?wind, voices, and adjacent court sounds. The system did not include multiple sensors; it relies solely on audio from a commercial microphone.

Sensors/Modalities Used

> "Using a single simple microphone placed in a non-invasive position and a notebook made it possible to detect the rally rhythm"

The system uses only **acoustic/audio input**?no accelerometers, cameras, or other modalities for the core detection task.

5. assistant (2026-06-06T17:34:31.293Z)

The key questions are: (1) what sport, (2) is the 85% genuinely a degradation of audio hit detection, and (3) is this a relevant, credible source. Let me dig deeper into the actual paper text to verify the WebFetch summary, since the summary itself drives the verdict.

6. user (2026-06-06T17:34:40.298Z)

Response

Journal and Publication Details:

This is a peer-reviewed article published in the *Journal of Functional Morphology and Kinesiology*, Volume 10, Issue 1, Article 47 (2025).

Quote containing "higher than 95%" and "85%":

"The ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%."

What each percentage measured:

- **95%:** Machine learning algorithm performance on detecting sound classifications (impacts and rebounds) in controlled testing conditions using cross-validation on the 1400-sample dataset
- **85%:** Overall system accuracy when applied during on-court groundstroke assessments, measuring both detection and correct timing measurements in real environmental conditions

On-court evaluation scope:

The on-court application involved 6 amateur players (3 female, 3 male) performing 128 total forehand shots across two groups (G1: 80 shots; G2: 48 shots) in structured drills on clay courts with environmental noise present.

On-court accuracy computation method:

"The system correctly recognized 78 impacts and 69 rebounds in G1, 48 impacts, and 40 rebounds in G2...The system correctly detected impacts in 96.25% of cases in G1 and 81.25% in G2 and rebounds in 86.25% of cases in G1 and 70.83% in G2."

7. assistant (2026-06-06T17:34:43.742Z)

I now have detailed numbers. Let me do one search to check the source quality and any contradicting context, and to confirm the publication is legitimate.

8. user (2026-06-06T17:34:49.509Z)

Web search results for query: "acoustic detection tennis ball impact rebound on-court 85% accuracy microphone Journal Functional Morphology Kinesiology 2025"

Links: [{"title": "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/"}, {"title": "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study | MDPI", "url": "https://www.mdpi.com/2411-5142/10/1/47"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis", "url": "https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"}, {"title": "[PDF] Detection of Tennis Events from Acoustic Data | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Detection-of-Tennis-Events-from-Acoustic-Data-Baughman-Morales/5757cc038c864dc0f2d676794faaa4ff8961a97f"}, {"title": "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11843912/"}, {"title": "Detection of Tennis Events from Acoustic Data | Proceedings of the 2nd International Workshop on Multimedia Content Analysis in Sports", "url": "https://dl.acm.org/doi/10.1145/3347318.3355520"}]

Based on the search results, I found the research you're looking for:

The article "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study" was published in the Journal of Functional Morphology and Kinesiology in 2025.

The acoustic detection system classifies the sound of the ball's impact on the racket and the ground to measure the time between them and give immediate feedback to the player. The ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%.

In the timing assessment of groundstrokes, the comparison with video analysis showed that the system correctly recognized 78 impacts and 69 rebounds in G1, 48 impacts, and 40 rebounds in G2, with the system correctly detecting impacts in 96.25% of cases in G1 and 81.25% in G2 and rebounds in 86.25% of cases in G1 and 70.83% in G2.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:35:02.573Z)

I have enough to render a verdict. Let me assess the claim against the evidence.

Claim: "The acoustic ML classifier achieves >95% detection accuracy, but the end-to-end on-court system drops to 85% average accuracy, showing real-world deployment degrades audio-only hit detection substantially."

Evidence assessment:

1. **Supported by quote?** The numeric core (95% classifier, 85% on-court system) is verbatim from the paper: "The ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%." The interpretive gloss ("real-world deployment degrades audio-only hit detection substantially") is a reasonable reading ? the paper attributes the drop to environmental noise on-court, and the system is audio-only (single microphone, no other modalities).

2. **Source quality:** Peer-reviewed (J. Functional Morphology and Kinesiology / MDPI, 2025, PMC-indexed). Current. However, it is explicitly a **pilot study**, the sport is **tennis** (not badminton), the on-court "system" includes timing measurement (impact?rebound interval) plus detection ? not pure hit-detection ? and the 85% is averaged over a very small sample (6 players, 128 shots). The per-group breakdown shows wide variance (impacts 96.25% in G1 vs 81.25% in G2; rebounds 86.25% vs 70.83%), indicating fragility but also that "85% average" is a coarse aggregate, not a robust generalizable figure.

Refutation analysis: The strong qualifier in the claim is "substantially" and "audio-only"

hit detection." Caveats: it's tennis not badminton (transferability caveat for the user's context), a small-n pilot, and the 85% bundles rebound (court-bounce) detection ? which tennis has but badminton does not ? plus timing accuracy, so it's not cleanly "hit detection." But these are scope/transfer caveats, not contradictions. The numbers are accurately quoted, the source is legitimate and current, and the qualitative point (audio detection degrades meaningfully in real on-court noise) is genuinely what the paper reports.

This survives refutation. The claim is well-supported and accurately sourced; the only weakness is that it's a small tennis pilot, which caps confidence at medium but does not refute it.

10. user (2026-06-06T17:35:11.136Z)

Structured output provided successfully

11. assistant (2026-06-06T17:35:17.599Z)

I've completed the verification and submitted my structured output.

Verdict: ****NOT refuted**** (confidence: medium).

The claim's numeric core is verbatim-supported by the primary peer-reviewed source (MDPI, 2025, PMC11843912): "The ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%." The system is genuinely audio-only (single microphone, no other sensors), and the paper attributes the on-court drop to environmental noise ? so the claim's interpretation is faithful.

Confidence is capped at medium because of scope/transfer caveats (not contradictions): it's a tennis pilot study with tiny n (6 players, 128 shots), not badminton, and the 85% bundles timing accuracy and tennis-specific rebound detection rather than being pure hit-detection. These limit generalizability to the user's badminton pipeline but don't refute the quoted claim.